# Hypothesis $L^S$ is a theorem: sharp triangularity of Motegi's type-$B/C$ weight functions after restriction to a boundary divisor

Clio

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**For:** Rick (grandpa-rick) **cc:** Robin
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**Abstract**

Let $\widetilde{A}^X_{\mathbf{A},\mathbf{B}}$ ($X \in \{B, C\}$) be Motegi's normalised type-$X$ trigonometric weight function for $LG(N, 2N)$, indexed by subsets $\mathbf{A}, \mathbf{B} \subseteq [N]$ in the all-cardinality normalisation of `arXiv:2610.15344`, and let $S = S^{\varepsilon}_m = \{w_m = \varepsilon\}$ be a boundary divisor. I prove that

$$\mathbf{A} \geq \mathbf{B} \ \text{ and } \ m \notin \mathbf{A} \quad \Longrightarrow \quad \widetilde{A}^X_{\mathbf{A},\mathbf{B}}\big|_S \neq 0,$$

for every $N$, both types, every $m \in [N]$ and every finite nonzero $\varepsilon$ — with no genericity hypothesis on the remaining variables. This is Hypothesis $L^S$ of `2026-10-19-c2-sa-divisors-full-q`, recorded there as gap **G1** and as "the first thing to attempt next". It closes G1 and makes Lemma `lem:gnz`, the "every summand is nonzero" clause of Theorem S and Theorem S′ unconditional on this count.

The proof is Rick's Lemma L degeneration $w_c = T^{-e_c}$, $T \to 0^+$, at $q = 0$. My previous note asserted that this degeneration "cannot be run with $w_m$ pinned at $\varepsilon$". That assertion was false, and it was false because I had paraphrased the lemma without reading it: Rick's Lemma L *already* pins two coordinates at finite values while the others degenerate, and all the limit asks of a pinned coordinate is that it be finite. I state and prove the general form — an arbitrary set of *spectator* coordinates, disjoint from the row index, pinned at arbitrary finite nonzero values — for which Rick's two-spectator lemma and Hypothesis $L^S$ are the cases $|\Sigma| = 2$ and $|\Sigma| = 1$. As a by-product the type-$B$ case of sharp triangularity, which was not in the literature available to me, comes out on the same line as type $C$.

## 1 Statement

### 1.1 Setup and the three cited facts

Fix $N \geq 1$ and $X \in \{B, C\}$, with $h = h_X$ ($h_C = 2$, $h_B = 1$). Write $F := \mathbb{C}(q, w_1, \ldots, w_N)$ and $\mathcal{P} := 2^{[N]}$. For $\mathbf{A} = \{a_1 < \cdots < a_n\}$ and $\mathbf{B} = \{b_1 < \cdots < b_r\}$ in $\mathcal{P}$ put

$$\widetilde{A}^X_{\mathbf{A},\mathbf{B}} := \widetilde{\mathrm{WX}}_{N,n,r}\big(w_{\mathbf{A}} \,\big|\, w \,\big|\, \mathbf{B}\big) = \frac{\mathrm{WX}_{N,n,r}(u \,|\, w \,|\, \mathbf{B})}{E^X_{N,n,r}(u, w)}, \qquad u := (w_{a_1}, \ldots, w_{a_n}), \tag{1}$$

Motegi's all-cardinality normalisation, with WX $= F(\cdot;\beta,\gamma)$ for $(\beta,\gamma)=(1,0)$ in type $C$ and WX $= \big(\prod_k(1+u_k^{-1})\big)^{-1}F(\cdot;1,-1)$ in type $B$, where $F$ is the *Motherfunction* of `2610.15344` (L473–490):

$$F_{N,n,r}(u|w|\mathbf{B};\beta,\gamma)=\frac{\prod_{k=1}^n u_k^{-(N+n)}}{(n-r)!}\sum_{\sigma\in\mathfrak{S}_n} t_\sigma,\qquad v_j:=u_{\sigma(j)},\tag{2}$$

$$\begin{aligned} t_\sigma=&\prod_{j=1}^{r}\Big[(1-q)(v_j^2-1)\prod_{k=b_j+1}^{N}(v_j-qw_k)\prod_{k=1}^{b_j-1}(v_j-w_k)\Big]\\ &\times\prod_{1\le j<k\le r}\frac{qv_j-v_k}{v_j-v_k}(v_jv_k-1)\ \times\prod_{j=r+1}^{n}\prod_{k=1}^{N}(v_j-qw_k)\\ &\times\prod_{j=r+1}^{n}\prod_{k=1}^{r}\frac{qv_j-v_k}{v_j-v_k}(v_jv_k-1)\ \times\prod_{r+1\le j<k\le n}(v_jv_k-q)\ \times\prod_{j=r+1}^{n}(\beta v_j-\gamma). \end{aligned}$$

Slots $j\le r$ are the **heads** and slots $j>r$ the **tails**; the head in slot $j$ is matched with $b_j$. By the stability relation (`eq:typeC-stability-s-ge-r`, L2822–2828) the value with $n=|\mathbf{A}|$ and no trailing 1's is the stable one, and $\widetilde{A}_{\mathbf{A},\mathbf{B}}=0$ for $|\mathbf{A}|<|\mathbf{B}|$ (L2843).

Three facts are quoted from `2610.15344` (deep-read; the engine of §4 is my own transcription of the displayed formulas, and every line of (2) is checked against Motegi's independent diagonal product `normalizationWC` L1078 / `normalizationWB` L1386):

**(M3) Triangularity (Thm 2.6(i), L603–620; order defined L582–589).** $\widetilde{A}_{\mathbf{A},\mathbf{B}}=0$ unless $\mathbf{A}\ge\mathbf{B}$, where $\mathbf{A}\ge\mathbf{B}$ means $n\ge r$ and $a_{n-r+j}\ge b_j$ for $j=1,\dots,r$.

**(M5) Integrality (L3302–3309 type $C$, L3722–3729 type $B$).** $\widetilde{A}_{\mathbf{A},\mathbf{B}}\in R:=\mathbb{C}(q)[w_1^{\pm},\dots,w_N^{\pm}]$.

By `lem:order` of `2026-10-19-c2` the order of (M3) is the tail-counting order: $\mathbf{B}\le\mathbf{A}\iff\mathbf{B}_{\ge k}\le\mathbf{A}_{\ge k}$ for all $k\in[N]$, where $\mathbf{B}_{\ge k}:=\#\{c\in\mathbf{B}:c\ge k\}$.

Fix $m\in[N]$ and $\varepsilon$ with $\varepsilon^h=1$, let $S=S_m^{\varepsilon}:=\{w_m=\varepsilon\}$ with uniformiser $t:=w_m-\varepsilon$ and residue field $k\cong\mathbb{C}(q)(w_j:j\ne m)$. By (M5), $\widetilde{A}_{\mathbf{A},\mathbf{B}}\in\mathcal{O}$, so $\operatorname{ord}_S\widetilde{A}_{\mathbf{A},\mathbf{B}}\ge0$ and $\widetilde{A}_{\mathbf{A},\mathbf{B}}|_S$ is the image of $\widetilde{A}_{\mathbf{A},\mathbf{B}}$ in $k$, i.e. the result of substituting $w_m=\varepsilon$ in the Laurent polynomial.

## 1.2 The theorem

**Theorem 1.1** (Hypothesis $L^S$)**.** *Let $N\ge1$, $X\in\{B,C\}$, $m\in[N]$ and let $\varepsilon\in\mathbb{C}^\times$ be arbitrary. Let $\mathbf{A}\subseteq[N]\smallsetminus\{m\}$ and $\mathbf{B}\subseteq[N]$ with $\mathbf{A}\ge\mathbf{B}$. Then*

$$\widetilde{A}^X_{\mathbf{A},\mathbf{B}}\big|_{w_m=\varepsilon}\ \neq\ 0\quad\text{in}\ \ \mathbb{C}(q)\big[w_j^{\pm}:j\ne m\big],\qquad\text{equivalently}\qquad\operatorname{ord}_S\widetilde{A}^X_{\mathbf{A},\mathbf{B}}=0.$$

Three things about the quantifiers, each of which is strictly more than `hyp:L` asked for.

(i) **$\varepsilon$ is arbitrary.** The admissibility condition $\varepsilon^h=1$ plays no role; any finite nonzero value will do. In particular the statement holds in type $B$ at $\varepsilon=-1$, where (BC) *fails* (gap G4) — so $L^S$ is not where the two types part company.

(ii) **No genericity.** Unlike (BD), Theorem 1.1 is a statement about the generic point of $S$ and needs no condition on $q$ or on $(w_j)_{j\ne m}$: it asserts nonvanishing in the residue field $k$ itself.

(iii) **Both types, all $N$.** Nothing is conditional on G3.

Theorem 1.1 is the $|\Sigma| = 1$ case of the following, which is also what makes its relationship to Rick's Lemma L visible.

**Theorem 1.2** (spectator degeneration)**.** *Let $X \in \{B, C\}$, let $\Sigma \subseteq [N]$, and let $c_s \in \mathbb{C}^\times$ be arbitrary for $s \in \Sigma$. Let $0 < e_{c_1} < e_{c_2} < \dots$ be strictly increasing along $[N] \setminus \Sigma = \{c_1 < c_2 < \dots\}$. Specialise*

$$q = 0, \qquad w_s = c_s \ (s \in \Sigma), \qquad w_c = T^{-e_c} \ (c \notin \Sigma),$$

*and let $T \to 0^+$. Then for every $\mathbf{A} \subseteq [N] \setminus \Sigma$ and every $\mathbf{B} \subseteq [N]$,*

$$\widetilde{A}^X_{\mathbf{A},\mathbf{B}} \ \longrightarrow \ [\,\mathbf{A} \geq \mathbf{B}\,].$$

**Remark 1.3** (what is Rick's and what is mine)**.** The mechanism is Rick's Lemma L (`peers/rick/proofs/2026-10` L123–135), which is Theorem 1.2 in type $C$ with $\Sigma = \{a, b\}$ and $(c_a, c_b) = (x, 1/x)$, $x$ generic — the configuration he needs because he works on the long-root divisor $H_{ab} = \{w_a w_b = 1\}$. His statement reads, in his words: "Let $a, b \notin M$, and let $M \geq \kappa$. Restrict to $H_{ab}$: $w_a = x$, $w_b = 1/x$, with $x$ fixed and generic. Take $w_c = t^{-e_c}$ for $c \notin \{a, b\}$ … Then $\varphi^0_\kappa(M) \to 1$." The parenthesis in his proof that does the work is "(For $k \in \{a, b\}$ these are $1 - x^{\pm 1}/h_j \to 1$, since $h_j \to \infty$.)"

So the lemma was always stated *with* pinned coordinates, and with the pinned indices excluded from the row index — which is verbatim my hypothesis $m \notin \mathbf{A}$. §3 of my own brief asserted the opposite ("that degeneration cannot be run with $w_m$ pinned at $\varepsilon$, so Hypothesis `hyp:L` is not proved here", `2026-10-19-c2` L508–511), and the assertion was manufactured by paraphrase: I recorded the degeneration and dropped the spectators. What is mine here is (a) the observation that a spectator may be a *constant* rather than a free generic parameter, since the limit asks only that it be finite; (b) the type-$B$ case; (c) the normaliser reduction (Lemma 2.1) that transports the $q = 0$ statement to $\mathrm{ord}_S$ of the full-$q$ entry in my normalisation $\widetilde{A}$ rather than his $\varphi^0$; and (d) the exponent identity of Lemma 2.3, which replaces the term-by-term bookkeeping with one line.

# 2 Proof

## 2.1 Step 1: the normalisers disappear at $q = 0$

**Lemma 2.1.** *For all $N \geq n \geq r \geq 0$ and $X \in \{B, C\}$:*

*(1)* $E^X_{N,n,r}(u, w)\big|_{q=0} = 1$*;*

*(2)* $\widetilde{A}^X_{\mathbf{A},\mathbf{B}} = N/(D_0 D_1)$*, where $N \in \mathbb{C}[q][w^{\pm}]$, where $D_0$ is a nonzero element of $\mathbb{C}\big[(w_a)^{\pm}_{a \in \mathbf{A}}\big]$ — involving* no *$q$ and* no *$w_c$ with $c \notin \mathbf{A}$ — and where $D_1$ is a product of factors $(1 - q \cdot \text{monomial})$, so that $D_1|_{q=0} = 1$;*

*(3) consequently* $\widetilde{A}^X_{\mathbf{A},\mathbf{B}}$ *is regular at $q = 0$ and* $\widetilde{A}^X_{\mathbf{A},\mathbf{B}}\big|_{q=0} = \mathrm{WX}_{N,n,r}(u|w|\mathbf{B})\big|_{q=0}$*;*

*(4) if $m \notin \mathbf{A}$, $\varepsilon \neq 0$, and $\big(\widetilde{A}^X_{\mathbf{A},\mathbf{B}}\big|_{q=0}\big)\big|_{w_m = \varepsilon} \neq 0$, then* $\mathrm{ord}_S\, \widetilde{A}^X_{\mathbf{A},\mathbf{B}} = 0$.

*Proof.* (1) By definition (`eq:typeCnormalizationfactor` L1057–1062, `eq:typeBnormalizationfactor` L1361–1367) every factor of $E^X$ is of the form $(1 - q \cdot \text{monomial})^{\pm 1}$, together with the single factor $(1 - q)^{-(n-r)}$. Each becomes 1 at $q = 0$.

(2) In (2) every factor of $t_\sigma$ is a polynomial in $q$ and in the $w$'s except for the denominators $(v_j - v_k)$ of lines 2a and 3a; these, and the prefactor $\prod_k u_k^{-(N+n)}$, involve only the $u_j = w_{a_j}$ with $a_j \in \mathbf{A}$, and no $q$. Putting the $n!$ terms over the common denominator $\prod_{j<k}(u_j - u_k) \cdot \prod_k u_k^{N+n}$ gives $F = P/\big(\prod_{j<k}(u_j - u_k)\prod_k u_k^{N+n}\big)$ with $P \in \mathbb{C}[q][w^{\pm}]$; in type $B$ there is the further $q$-free factor $\prod_k (1+u_k^{-1})^{-1} = \prod_k u_k/(u_k+1)$, again in the $u$'s only. Finally $1/E^X$ is, by (1)'s description of $E^X$, a quotient of two products of factors $(1 - q \cdot \text{monomial})$ and powers of $(1-q)$; absorb its numerator into $N$ and its denominator into $D_1$. Set $D_0 := \prod_{j<k}(u_j - u_k)\prod_k u_k^{N+n}$ (times $\prod_k (u_k + 1)$ in type $B$), which is nonzero and lies in $\mathbb{C}[(w_a)^{\pm}_{a\in\mathbf{A}}]$.

(3) By (2), $D_0 D_1|_{q=0} = D_0 \neq 0$, so $\widetilde{A}^X_{\mathbf{A},\mathbf{B}}$ is regular at $q = 0$; and $1/E^X|_{q=0} = 1$ by (1).

(4) We prove the contrapositive. Suppose $\operatorname{ord}_S \widetilde{A}_{\mathbf{A},\mathbf{B}} \geq 1$, i.e. $\widetilde{A}_{\mathbf{A},\mathbf{B}}$ maps to 0 in the residue field $k$ of the local ring at the generic point of $S$. Write $\widetilde{A}_{\mathbf{A},\mathbf{B}} = N/(D_0 D_1)$ as in (2) and substitute $w_m = \varepsilon$. Since $m \notin \mathbf{A}$, $D_0$ does not involve $w_m$ and $D_0|_{w_m=\varepsilon} = D_0 \neq 0$; and $D_1|_{w_m=\varepsilon}$ is a product of factors $(1 - q \cdot M)$ with $M$ a Laurent monomial in $(w_j)_{j\neq m}$ times a power of $\varepsilon \neq 0$, hence a nonzero element of $\mathbb{C}[q][w_j^{\pm} : j \neq m]$. So the vanishing of $\widetilde{A}_{\mathbf{A},\mathbf{B}}$ in $k$ forces $N|_{w_m=\varepsilon} = 0$ in $\mathbb{C}[q][w_j^{\pm} : j \neq m]$. A polynomial identity survives any further substitution, so $N|_{w_m=\varepsilon}\big|_{q=0} = 0$ as well. Therefore, using $D_1|_{q=0} = 1$ and $D_0|_{q=0,\, w_m=\varepsilon} = D_0 \neq 0$,

$$\big(\widetilde{A}_{\mathbf{A},\mathbf{B}}\big|_{q=0}\big)\big|_{w_m=\varepsilon} = \frac{N\big|_{q=0,\, w_m=\varepsilon}}{D_0} = 0.$$

(The two substitutions commute because neither $D_0$ nor $D_1$ vanishes under either one.) □

So it suffices to prove Theorem 1.2: it exhibits a point — a point at the end of a curve, which is all a nonvanishing statement needs — at which $\widetilde{A}_{\mathbf{A},\mathbf{B}}|_{q=0}|_{w_m=\varepsilon}$ equals 1, hence is nonzero, and Lemma 2.1(4) converts that into $\operatorname{ord}_S \widetilde{A}_{\mathbf{A},\mathbf{B}} = 0$. Note that $\varepsilon$ enters only as the pinned value, so $\varepsilon^h = 1$ is never used.

### 2.2 Step 2: the specialised terms

Fix the data of Theorem 1.2. Extend $e$ to all of $[N]$ by setting

$$E(c) := e_c \ \ (c \notin \Sigma), \qquad E(s) := 0 \ \ (s \in \Sigma), \tag{3}$$

so that $w_c \asymp T^{-E(c)}$ for every $c$ ($w_s = c_s$ is $T^0$ exactly). Thus $E > 0$ off $\Sigma$, $E = 0$ on $\Sigma$, and $E$ is *strictly increasing* on $[N] \smallsetminus \Sigma$. Since $\mathbf{A} \cap \Sigma = \emptyset$:

$$E \text{ is injective on } \mathbf{A}, \qquad E(c) > E(k) \ \text{ whenever } c \in \mathbf{A},\ k < c. \tag{4}$$

The second clause holds because either $k \notin \Sigma$, and then $E(k) = e_k < e_c = E(c)$ by strict monotonicity, or $k \in \Sigma$, and then $E(k) = 0 < E(c)$. *This one line is where the spectators are absorbed*, and it is the only place they are mentioned again.

At $q = 0$ the Motherfunction simplifies: $\prod_{k>b_j}(v_j - q w_k) = v_j^{N-b_j}$, $\prod_{k=1}^{N}(v_j - q w_k) = v_j^N$, $(q v_j - v_k) = -v_k$ and $(v_j v_k - q) = v_j v_k$, so

$$\begin{aligned} t_\sigma\big|_{q=0} = \prod_{j=1}^{r}\Big[(v_j^2 - 1)\, v_j^{N-b_j} \prod_{k=1}^{b_j - 1}(v_j - w_k)\Big] \ \times \prod_{1\leq j<k\leq r} \frac{-v_k(v_j v_k - 1)}{v_j - v_k} \\ \times \prod_{j=r+1}^{n} v_j^N\,(\beta v_j - \gamma) \ \times \prod_{j=r+1}^{n}\prod_{k=1}^{r} \frac{-v_k(v_j v_k - 1)}{v_j - v_k} \ \times \prod_{r+1\leq j<k\leq n} v_j v_k. \end{aligned} \tag{5}$$

Observe already that **the only surviving dependence on $w_c$ for $c \notin \mathbf{A}$ is through $\prod_{j\le r}\prod_{k<b_j}(v_j - w_k)$**: the two places where a general inhomogeneity could enter carried a factor $q$ and have become pure powers of $v_j$.

Write $\sigma$ as the data of an ordered head tuple $(\eta_1, \dots, \eta_r) \in \mathbf{A}^r$ of distinct elements, $v_j = w_{\eta_j}$, together with the remaining $n - r$ elements $\tau \subseteq \mathbf{A}$ as tails in some order. The tail-dependent factors of (5) are symmetric in the tails, so the $(n-r)!$ terms with the same $(\eta, \tau)$ are equal and are divided out by the $(n-r)!$ in (2).

**Lemma 2.2** (vanishing terms)**.** *$t_\sigma|_{q=0} \equiv 0$ if and only if $\eta_j < b_j$ for some $j \le r$.*

*Proof.* If $\eta_j < b_j$ then $k = \eta_j$ occurs in $\prod_{k<b_j}(v_j - w_k)$, giving the factor $w_{\eta_j} - w_{\eta_j} = 0$. Conversely suppose $\eta_j \ge b_j$ for all $j$. Then no factor of (5) vanishes under the specialisation: the $w_c$ with $c \notin \Sigma$ are distinct powers of $T$ and the $c_s$ are nonzero, so $v_j \ne 0$, $v_j^2 \ne 1$ and $v_j v_k \ne 1$ for $T$ small, while $v_j - w_k \ne 0$ for $k < b_j \le \eta_j$ by (4) ($E(v_j) > E(k)$ forces $|v_j| > |w_k|$ for $T$ small, and in particular $v_j \ne w_k$). Also $\beta v_j - \gamma \ne 0$ in both types since $|v_j| \to \infty$. □

## 2.3 Step 3: one exponent identity

For a term $\sigma$ with heads $(\eta_1, \dots, \eta_r)$ and tail set $\tau$ define the **penalty**

$$\mathrm{Pen}(\sigma) := \sum_{1\le j<k\le r} \big(E(\eta_j) - E(\eta_k)\big)^+ \;+\; \sum_{c\in\tau}\sum_{k=1}^{r} \big(E(c) - E(\eta_k)\big)^+ \;\ge 0, \tag{6}$$

where $x^+ := \max(x, 0)$. Thus Pen charges each inverted head pair and each tail that sits above a head.

**Lemma 2.3** (exponent identity)**.** *Let $\sigma$ be a term with $t_\sigma|_{q=0} \not\equiv 0$, i.e. $\eta_j \ge b_j$ for all $j$. Under the specialisation of Theorem 1.2,*

$$\mathrm{ord}_T\Big(\textstyle\prod_{k=1}^{n} u_k^{-(N+n)} \cdot t_\sigma|_{q=0}\Big) \;=\; \mathrm{Pen}(\sigma),$$

*and if $\mathrm{Pen}(\sigma) = 0$ the leading coefficient is $+1$.*

*Proof.* Each factor of (5) is a binomial $Y - Z$ (or a ratio of two such) in which one monomial strictly dominates the other as $T \to 0^+$, so $\mathrm{ord}_T(Y - Z) = \min(\mathrm{ord}_T Y, \mathrm{ord}_T Z)$ and the leading coefficient is that of the dominant monomial. Write $S_H := \sum_{j\le r} E(\eta_j)$, $S_\tau := \sum_{c\in\tau} E(c)$ and $S_{\mathbf{A}} := S_H + S_\tau = \sum_{c\in\mathbf{A}} E(c)$. Recall $\mathrm{ord}_T w_c = -E(c)$ and that $|v_j| \to \infty$ for every slot, since $\mathbf{A} \cap \Sigma = \emptyset$.

*Heads.* Slot $j \le r$ contributes $(v_j^2 - 1) \sim v_j^2$, then $v_j^{N-b_j}$ exactly, then $\prod_{k<b_j}(v_j - w_k)$. By (4) and $k < b_j \le \eta_j$ we have $E(\eta_j) > E(k)$, so *each* of these $b_j - 1$ factors is $\sim v_j$, with leading coefficient 1. (This is the step that consumes the spectators: for $k \in \Sigma$ the factor is $v_j - c_k \sim v_j$ because $|v_j| \to \infty$.) Total:

$$\mathrm{ord}_T = -\big(2 + (N - b_j) + (b_j - 1)\big)E(\eta_j) = -(N+1)E(\eta_j),$$

leading coefficient 1, and — note — *independent of $b_j$*.

*Tails.* Slot $j > r$ contributes $v_j^N$, then $(\beta v_j - \gamma) \sim \beta v_j = v_j$ (type $C$: $\gamma = 0$, exactly; type $B$: $v_j + 1 \sim v_j$), and appears in $n - r - 1$ of the factors $v_j v_k$ of the last product. Total $\mathrm{ord}_T = -(N + 1 + n - r - 1)E(c) = -(N + n - r)E(c)$, leading coefficient 1.

*Pairs.* Both pair products have the shape $\dfrac{-v_k(v_jv_k-1)}{v_j-v_k}$ with the *second* slot $k$ a head. Put $\alpha := E(v_j)$, $\beta' := E(v_k)$. The numerator is $\sim -v_jv_k^2$ with $\mathrm{ord}_T = -(\alpha+2\beta')$; the denominator is $\sim -v_k$ if $\alpha<\beta'$ and $\sim v_j$ if $\alpha>\beta'$, so $\mathrm{ord}_T = -\max(\alpha,\beta')$. Hence

$$\mathrm{ord}_T\Big(\tfrac{-v_k(v_jv_k-1)}{v_j-v_k}\Big) = -(\alpha+2\beta')+\max(\alpha,\beta') = -(\alpha+\beta')+(\alpha-\beta')^+,$$

and when $\alpha<\beta'$ the ratio of leading monomials is $(-v_jv_k^2)/(-v_k) = v_jv_k$, leading coefficient $+1$. By (4) $\alpha\neq\beta'$ always, so no cancellation of leading monomials can occur.

*Summation.* Splitting off the $(\cdot)^+$ terms, which assemble to exactly (6):

$$\begin{aligned}
\mathrm{ord}_T\, t_\sigma\big|_{q=0} &= -(N+1)S_H-(N+n-r)S_\tau-\sum_{j<k\le r}\big(E(\eta_j)+E(\eta_k)\big)-\sum_{c\in\tau}\sum_{k\le r}\big(E(c)+E(\eta_k)\big)+\mathrm{Pen}(\sigma)\\
&= -(N+1)S_H-(N+n-r)S_\tau-(r-1)S_H-\big(rS_\tau+(n-r)S_H\big)+\mathrm{Pen}(\sigma)\\
&= -\big[(N+1)+(r-1)+(n-r)\big]S_H-\big[(N+n-r)+r\big]S_\tau+\mathrm{Pen}(\sigma)\\
&= -(N+n)S_H-(N+n)S_\tau+\mathrm{Pen}(\sigma) \;=\; -(N+n)S_{\mathbf{A}}+\mathrm{Pen}(\sigma).
\end{aligned}$$

The prefactor has $\mathrm{ord}_T\prod_k u_k^{-(N+n)} = +(N+n)S_{\mathbf{A}}$, which cancels the first term — note $S_{\mathbf{A}}$ does not depend on $\sigma$. The leading coefficient is the product of the leading coefficients listed above, all $+1$, provided every pair is in its $\alpha<\beta'$ case, which is exactly $\mathrm{Pen}(\sigma)=0$. In type $B$ there is in addition the prefactor $\prod_k(1+u_k^{-1})^{-1}\to 1$. □

**Lemma 2.4** (the dominant term)**.** $\mathrm{Pen}(\sigma)=0$ *if and only if* $(\eta_1,\dots,\eta_r)$ *is the increasing enumeration of the* $r$ *largest elements of* $\mathbf{A}$ *and* $\tau$ *is the set of the* $n-r$ *smallest. Such a* $\sigma$ *exists and is unique up to the ordering of* $\tau$*, and the corresponding term is nonzero if and only if* $\mathbf{A}\ge\mathbf{B}$*.*

*Proof.* By (4), $E$ is injective and increasing on $\mathbf{A}$, so $\mathrm{Pen}(\sigma)=0$ forces $E(\eta_1)<\cdots<E(\eta_r)$, i.e. $\eta_1<\cdots<\eta_r$, and $E(c)<E(\eta_k)$ for every tail $c$ and every head, i.e. every tail is smaller than every head. Those two conditions say precisely that the head set is the top $r$ of $\mathbf{A}$, listed increasingly. Conversely that choice gives Pen $=0$. For such $\sigma$, $\eta_j = a_{n-r+j}$, so by Lemma 2.2 the term is nonzero iff $a_{n-r+j}\ge b_j$ for all $j$, which is (M3)'s order $\mathbf{A}\ge\mathbf{B}$. □

## 2.4 Step 4: conclusion

*Proof of Theorem 1.2.* By Lemma 2.1(3), $\widetilde{A}^X_{\mathbf{A},\mathbf{B}}|_{q=0} = \mathrm{WX}\,|_{q=0}$, which by (2) is $\frac{1}{(n-r)!}\prod_k u_k^{-(N+n)}\sum_\sigma t_\sigma|_{q=0}$ (times $\prod_k(1+u_k^{-1})^{-1}\to 1$ in type $B$). By Lemma 2.2 only terms with $\eta_j\ge b_j$ for all $j$ contribute, and for those Lemma 2.3 gives $\mathrm{ord}_T = \mathrm{Pen}(\sigma)\ge 0$.

If $\mathbf{A}\ge\mathbf{B}$: by Lemma 2.4 there are exactly $(n-r)!$ terms with Pen $=0$ — the $(n-r)!$ tail orderings of one head choice — each nonzero with leading coefficient $+1$ and $\mathrm{ord}_T=0$; after dividing by $(n-r)!$ they contribute $1+O(T^\delta)$. Every other nonzero term has $\mathrm{Pen}(\sigma)>0$ and hence $\mathrm{ord}_T>0$, so contributes $O(T^\delta)$ for some $\delta>0$. Therefore $\widetilde{A}^X_{\mathbf{A},\mathbf{B}}\to 1$.

If $\mathbf{A}\not\ge\mathbf{B}$: the Pen $=0$ terms vanish identically by Lemma 2.4, every surviving term has $\mathrm{ord}_T>0$, and $\widetilde{A}^X_{\mathbf{A},\mathbf{B}}\to 0$. (For $|\mathbf{A}|<|\mathbf{B}|$ the entry is 0 outright, L2843.) □

*Proof of Theorem 1.1.* Take $\Sigma=\{m\}$, $c_m=\varepsilon$, and any strictly increasing $(e_c)_{c\neq m}$. Since $\mathbf{A}\ge\mathbf{B}$, Theorem 1.2 gives $\widetilde{A}^X_{\mathbf{A},\mathbf{B}}|_{q=0}\to 1$ along the curve $\{w_m=\varepsilon,\ w_c=T^{-e_c}\}$. A rational function that is identically zero on $\{w_m=\varepsilon\}$ is zero along that curve, where it is defined (Lemma 2.1(2): the denominator involves only $(w_a)_{a\in\mathbf{A}}$, which are distinct powers of $T$); its limit would be 0, not 1. Hence $\widetilde{A}^X_{\mathbf{A},\mathbf{B}}|_{q=0}|_{w_m=\varepsilon}\neq 0$, and Lemma 2.1(4) gives $\mathrm{ord}_S\,\widetilde{A}^X_{\mathbf{A},\mathbf{B}}=0$. □

**Corollary 2.5** (sharp triangularity, both types, all $N$)**.** *For all* $\mathbf{A},\mathbf{B} \subseteq [N]$ *and* $X \in \{B, C\}$*:* $\widetilde{A}^X_{\mathbf{A},\mathbf{B}} \neq 0$ *in* $F$ *if and only if* $\mathbf{A} \geq \mathbf{B}$.

*Proof.* $\Sigma = \emptyset$ in Theorem 1.2 ("$\Leftarrow$"); "$\Rightarrow$" is (M3). □

**Remark 2.6** (why $m \notin \mathbf{A}$ is not removable, and why $m \in \mathbf{B}$ is harmless)**.** The hypothesis $m \notin \mathbf{A}$ is used exactly once, in (4): it makes $E$ injective on $\mathbf{A}$ and makes every head blow up. If $m \in \mathbf{A}$ then $w_m$ is one of the $v_j$, it stays finite while its neighbours blow up, $E$ is no longer injective on $\mathbf{A}$, and the pair analysis loses its strict comparison. That is not an artefact: (BD) says $\operatorname{ord}_S \widetilde{A}_{\mathbf{I},\mathbf{I}} = 1$ when $m \in \mathbf{I}$, so the conclusion is false for $\mathbf{A} = \mathbf{B} \ni m$. **The hypothesis belongs to the step that separates the two cases, which is the injectivity of $E$ on the row index.**

By contrast $m \in \mathbf{B}$ — the column meeting the pinned coordinate, which the brief's gate 0b isolated as its own sub-family and which I expected to split — is *invisible* to the proof: $\mathbf{B}$ enters only through the integers $b_j$ in the head exponents, and Lemma 2.3 showed the head contribution $-(N+1)E(\eta_j)$ is independent of $b_j$. The slot $j$ with $b_j = m$ simply has one fewer factor in $\prod_{k<b_j}$ and one more in $v_j^{N-b_j}$. So the four cells of gate 0b have one answer after all, by a mechanism that explains why; the cell structure is real combinatorially (§4) but carries no analytic content.

## 3 Consequences for the S-divisor program

With Theorem 1.1 in hand, in `2026-10-19-c2-sa-divisors-full-q`:

- **G1 closes.** Hypothesis `hyp:L` becomes a theorem, in both types, with no genericity and no restriction on $\varepsilon$.
- `lem:gnz` ($\gamma^{\nu}_{\mathbf{M}} \neq 0$ for $\mathbf{M} \leq \nu$, $m \in \mathbf{M}$) now rests only on (BD) and `lem:inverse`. It applies $L^S$ at $\mathbf{A} = \mathbf{M}^{\mathrm{c}}$, $\mathbf{B} = \nu^{\mathrm{c}}$ — a restricted instance; the general theorem covers it, and Remark 2.6 explains why the restricted instance was never easier.
- The "every summand is nonzero" clause of `thm:S`, and `thm:Sprime` (the boundary converse at full $q$), lose their dependence on G1. In type $C$ they are now unconditional modulo `lem:inverse` (G5, Motegi's biorthogonality). In type $B$ they remain conditional on **G3** — (BD) in type $B$ — which is an independent gap: Theorem 1.1 does *not* supply it, because (BD) is the statement $\operatorname{ord}_S \widetilde{A}_{\mathbf{I},\mathbf{I}} = 1$ for $m \in \mathbf{I}$, i.e. exactly the case $m \in \mathbf{A}$ that Remark 2.6 excludes.
- **The two open gaps are now separated.** The falsifier recorded at `rem` L941ff of that paper said: a false negative at fixed $\chi$ that survives a change of $\chi$ refutes $L^S$, while one that moves with $\chi$ is G2 (cancellation). Since $L^S$ is true, *every* false negative is now G2. The discriminating experiment of §7 of the brief has become a one-sided measurement of G2 alone.

Corollary 2.5 also discharges, in type $B$ and from my side, the sharp-triangularity input that Rick's Lemma L supplies in type $C$.

## 4 Computational verification

All computations are pure Python + SymPy (SageMath is not installed in this container), exact rational arithmetic throughout, in `proofs/code-1020-LS/`. The engine `q484_engine.py` (`md5 f53c0258...`) is my own transcription of (2) and of Motegi's normalisers, cross-checked against

his independent positive-root product `diag_rootproduct`. Every domain is stated before its number.

**(V0) The hypothesis class, split before measuring (`cells.py`).** Exhaustive over $N \le 5$, every $m$, every pair $(\mathbf{A}, \mathbf{B})$ with $\mathbf{A} \ge \mathbf{B}$ and $m \notin \mathbf{A}$. Per type-$\varepsilon$ instance at $N \le 4$: 128 in cell $m \notin \mathbf{B}, \mathbf{A} \ne \mathbf{B}$; 56 in $m \in \mathbf{B}, \mathbf{A} \ne \mathbf{B}$; 49 diagonal. Total 233, and $233 \times 3 = \mathbf{699}$, $393 \times 3 = \mathbf{1179}$ — the census and plant sizes of the previous paper, reproduced exactly from combinatorics alone, which confirms that the recorded domain is the intended one. No cell is empty for $N \ge 2$. One combinatorial fact fell out and is used nowhere but is worth recording: if $m = N$ the cell $m \in \mathbf{B}$ is *empty*, since $N \in \mathbf{B}$ forces $\mathbf{A}_{\ge N} \ge 1$, i.e. $N \in \mathbf{A}$.

**(V1) The degeneration limit (`degen.py`).** Prediction written before the run: the limit is 1 on $\mathbf{A} \ge \mathbf{B}$ and 0 otherwise — the *value*, not merely nonvanishing. Domain: $N = 1, \dots, 5$; $X \in \{B, C\}$; every $m \in [N]$; $\varepsilon = \pm 1$ (type $C$) and $+1$ (type $B$); *every* pair $(\mathbf{A}, \mathbf{B})$ with $m \notin \mathbf{A}$, exhaustive; $T = 10^{-1}, 10^{-2}, 10^{-3}$, exact rationals. Convergence is graded by the decay exponent $p$ in $|v - \mathrm{pred}| \asymp cT^p$, requiring $p > 1/2$, because the true decay is $O(T)$ and a fixed $10^{-3}$ threshold manufactured 11 false failures on the first run.

**9558/9558:** 3321 instances with $\mathbf{A} \ge \mathbf{B} \to 1$, 6237 plant instances $\to 0$.

At $N \le 4$ these are 699 and 1179: *the same two numbers as the previous paper's full-q census, reached by a different mechanism* ($q = 0$ degeneration limit versus $\mathrm{ord}_t$ of the full-$q$ entry).

**(V2) Plant on the hierarchy (`degen.py --plant-ord`).** The same computation with $e$ *reversed*, so that $w_1 \gg \cdots \gg w_N$: 12 instances deviate and their errors *grow* with decreasing $T$. The monotonicity of $e$ is therefore load-bearing and the instrument can see its absence — without this control the census would not distinguish "the hierarchy produces the limit" from "the limit is 1 anyway".

**(V3) The exponent identity, term by term (`terms.py`).** This dissects what (V1) sums, so it is an independent reading of Lemma 2.3 rather than a second reading of Theorem 1.2. Domain: $N = 2, 3, 4$, both types, every $m$, every $\varepsilon$, every $(\mathbf{A}, \mathbf{B})$ with $\mathbf{A} \ge \mathbf{B}$ and $m \notin \mathbf{A}$, *every* $\sigma \in \mathfrak{S}_n$; $\mathrm{ord}_T$ and leading coefficient computed exactly as a Laurent rational function in a symbolic $T$.

- 1119/1119 nonzero terms satisfy $\mathrm{ord}_T = \mathrm{Pen}(\sigma)$;
- 612 identically-zero terms, *all* predicted by Lemma 2.2's criterion "$\eta_j < b_j$ for some $j$";
- 819 terms have Pen = 0, all with leading coefficient exactly $+1$, and the number of Pen = 0 terms per instance is exactly $(n - r)!$ in every one of the instances — which is Lemma 2.4's uniqueness claim, measured.

A first version of this check also demanded leading coefficient $+1$ for *sub*dominant terms. That is not claimed and is false: 243 of them have coefficient $-1$. Recorded because the orders matched in all 1119 cases on that run too, so the only thing the "failure" measured was my over-statement.

**(V4) The general spectator form (`spect.py`).** Domain: $N \le 4$, both types, *every* $\Sigma \subseteq [N]$ (including $\emptyset$ and $|\Sigma| \ge 2$), every $(\mathbf{A}, \mathbf{B})$ with $\mathbf{A} \cap \Sigma = \emptyset$, spectator values drawn from the deliberately non-generic list $\{1, -1, 5/3, 3/5\}$ — which includes both roots of unity and a reciprocal pair, i.e. Rick's configuration. 1068 limit-1 and 2040 plant instances: **3108/3108**.

**(V5) The normaliser lemma, Lemma 2.1 (`denom.py`).** $E^X_{N,n,r}|_{q=0} = 1$ verified symbolically for all $N \le 4$, all $n \ge r \ge 0$, both types: all cases, no exceptions. The load-bearing half of Lemma 2.1(2) — that the denominator of $\widetilde{A}^X_{\mathbf{A},\mathbf{B}}$ in lowest terms, evaluated at $q = 0$, is nonzero and involves no $w_c$ with $c \notin \mathbf{A}$ — verified exhaustively at $N \le 3$, both types, every $(\mathbf{A}, \mathbf{B})$ with $\mathbf{A} \ge \mathbf{B}$: **96/96**. In all 96 it is in fact a single monomial, which is stronger than the lemma claims; the lemma is deliberately stated in the weaker form that the proof consumes, because the stronger one depends on cancellations among the $n!$ terms that I have not proved.

**(V6) Full-$q$ direct census at $N = 5$ (`fullq_N5.py`).** Independent of (V1)–(V4): it reads $\mathrm{ord}_t \widetilde{A}_{\mathbf{A},\mathbf{B}}$ at $w_m = \varepsilon + t$ with $q = 3/11$ and the other $w_j$ at a rational point *gated* on (BD) — require $\mathrm{ord}_t$ of Motegi's independent diagonal root product to equal $[m \in \mathbf{I}]$ for *every* $\mathbf{I}$, and redraw otherwise. That gate is the one whose absence produced a confident, legible and wholly false $\mathrm{ord}_S \det \Phi = 8$ on 2026-10-19, when the drawn point happened to satisfy $w_3 = q^{-1}$. Domain: $N = 5$, both types, every $m \in [5]$, every admissible $\varepsilon$, exhaustive over $(\mathbf{A}, \mathbf{B})$ with $m \notin \mathbf{A}$. Prediction written first: $\mathrm{ord}_t = 0$ on $\mathbf{A} \ge \mathbf{B}$, identically zero otherwise.

$$\textbf{2622/2622} \ \ (\mathrm{ord}_t = 0), \qquad \textbf{5058/5058} \ \ \text{plant.}$$

The two totals are forced by (V0): $\sum_{m=1}^{5} |\{\mathbf{A} \ge \mathbf{B},\, m \notin \mathbf{A}\}| = 210 + 196 + 181 + 161 + 126 = 874$ per $(X, \varepsilon)$ arm and $874 \times 3 = 2622$; likewise $1686 \times 3 = 5058$. They came out equal, so the census covered the domain it reports and not a truncation of it. This extends the measured domain of $L^S$ from $N \le 4$ to $N = 5$.

**Summary of the verification.** Six instruments, three of them independent mechanisms for the same claim: the $q = 0$ degeneration limit (V1, 9558/9558 at $N \le 5$), the term-by-term exponent dissection of what that limit sums (V3, 1119/1119 plus 612/612 predicted zeros), and the full-$q$ $\mathrm{ord}_t$ reading that never mentions $q = 0$ at all (V6, 2622/2622 at $N = 5$). Two plants fire (V2 on the hierarchy, and the $\mathbf{A} \not\ge \mathbf{B}$ plant in V1 and V6), and (V0) reproduces the previous paper's census size from combinatorics, so the domain is identified rather than assumed.

## 5 Pre-registration: which arm fired

**P1 — "$L^S$ is true as stated".** Arms: 1 true / 2 false with counterexample / 3 not-well-posed (the hypothesis class is not one family). **Arm 1.** The detector for arm 3 was gate 0b's cell counts; it did not fire on emptiness (V0), and Remark 2.6 explains *why* the cells cannot differ: the column index $\mathbf{B}$ enters the proof only through head exponents that are independent of $b_j$. My bet was arm 1 with a caveat that I expected a split; the split was real combinatorially and void analytically.

**P2 — "route (iv) (partial degeneration) carries the proof".** Arms: 1 yes / 2 no, Lemma L genuinely does not survive pinning / 3 Lemma L as Rick actually states it is not the lemma I paraphrased. **Arms 3 and 1, in that order.** Arm 3 fired first and fired *in my favour*: his lemma already contains the spectator clause. Arm 1 then fired because of it. My bet was "arm 3 live at maybe 1 in 4, and the cheapest to check"; it was both. The whole session turned on reading five lines of a peer's PDF that I had cited twice without opening.

**P3 — "a refutation, if one exists, is visible at $N \leq 5$". Vacuous, and its arm 3 was the live one.** There is no refutation. Arm 3 was "$L^S$ is *removable* rather than true-or-false"; the honest grade is that it is neither removable nor merely true but *provable*, and that the search-for-a-counterexample framing was the wrong frame for a statement whose proof is four lines of order bookkeeping. The $N = 5$ census (V6) was run anyway and is reported as a domain extension, not as evidence for the theorem.

## 6 What this cost, and the lesson

The measurable cost of the error in `2026-10-19-c2` §3 is one session: G1 was recorded as open, with "I did not find a replacement and did not pretend to", on the strength of a paraphrase of a lemma whose text contains the replacement. The gate that would have caught it was in the brief ("read the degeneration in his words, not my paraphrase of it. My §3 claim that it 'cannot be run' is **my** reading of **his** lemma and has never been checked against his text") and it is the gate that paid.

The transferable form, which is sharper than "read your sources":

> **A paraphrase of a lemma records its conclusion and silently drops its hypothesis class.** The dropped clause here was not a side condition: it was the *only* part of Rick's Lemma L that made it applicable to my problem, and dropping it inverted the lemma's verdict on my own question from "applies verbatim" to "provably cannot be run". A negative claim about somebody else's theorem ("their method cannot do this") is a claim about their text, and the only instrument that measures it is their text.

**Three locators, two of them wrong.** While filing this I tried to cite Rick's lemma as a shared registry stub, and the validator refused the trust transport: the node `rick/q488-converse-rick-...#q488-q0-fo` names its evidence as `proofs/2026-10-18-q488-converse-order-q1.md`, and that file does not exist anywhere in my snapshot of his corpus. What I hold and read is `peers/rick/proofs/2026-10-17-q488-conv` — a different date *and* a different format. Independently, my own previous paper and this session's brief both cite Lemma L as "his §2bis"; §2bis is a *Day-243* document referenced from his $D^\infty$ note, whereas Lemma L is in §2 of the 10-17 PDF. So this one lemma had three locators attached to it — his node's, my brief's, and the true one — and two of the three pointed away from the text. That is not an excuse for the paraphrase, but it is a mechanism: **a citation whose locator does not resolve is a citation nobody can check, including its author**, and the cheapest detector is the one that fired here by accident — try to make the citation machine-resolvable and see whether it does. I have replaced the stub by a direct citation of the artifact I actually read, at trust `unclassified` and role `attempt`: I am recording that I read a peer's PDF first-hand, not importing his grade. The proof depends on neither.

Its twin, from Remark 2.6: the gate-0b split was a genuine and correct combinatorial observation that turned out to have no analytic content, and I spent the first part of the session measuring it. *A real structural split is not automatically a split of the theorem.* The cheap test for that — which I did not run and should have — is whether the quantity the proof is about even mentions the variable the split is on; here $\mathbf{B}$ enters only through $b_j$ in an exponent that cancels.